

Lecture 1

Introduction and Fundamentals

Elena Raponi | Neural Computing course



Universiteit
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The Netherlands

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3. Biological Background
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Presentation of the course

Course objectives

The course gives a **comprehensive overview of the field** with a series of lectures and programming exercises.

Key objectives:

1. Understand the fundamental principles of neural computation.
2. Learn the major neural network models, e.g., convolutional and recurrent NNs.
3. Practice implementing the NN models with PyTorch.
4. Get familiar with the computational tasks that are commonly tackled by NNs, e.g., image classification.

Course Organization

- **Time:** Every Wednesday, 15:15 – 17:00
- **Location:** Gorlaeus Building – CM.1.26
- **Lecturer:** Elena Raponi (e.raponi@liacs.leidenuniv.nl)
Office: Gorlaeus Building
Einsteinweg 55
2333 CC Leiden
Room number BE.4.19
- **Teaching Assistants:** Valeria Rezan, David Wünsch, Ivan Olarte Rodriguez, Noam Dunskey, Viktoria Georgieva, Nicolas Ramos Fernandez, and Daria Semenova.
- **Office Hours:** By appointment, please email directly the lecturer.
- **Content-related questions:** Make use of the Brightspace Forum!

Important Dates

1. Exam08/06/2026
2. Retake30/06/2026
3. Project Assignment15/04/2026
4. Project Deadline15/05/2026

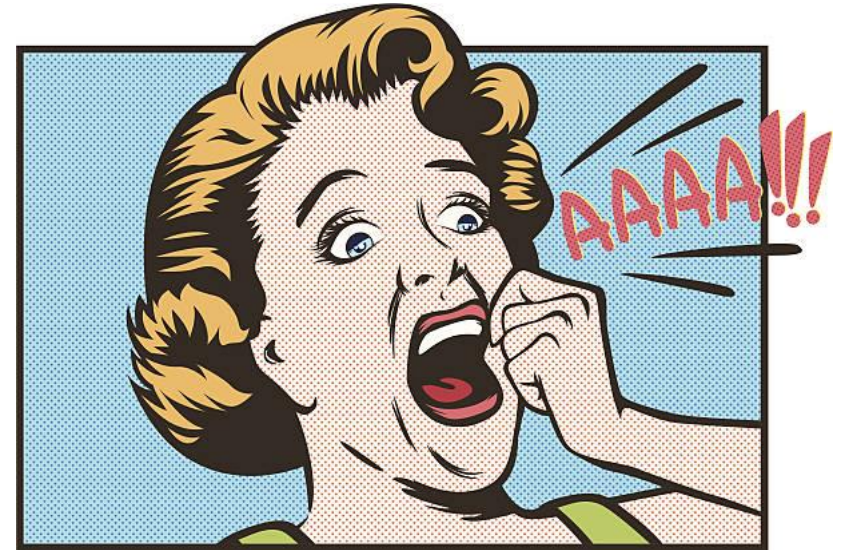
Grading Policy

The final grade is a combination of grades for (1) the written exam (70%) and (2) the practical assignment (30%).

Both the written exam and the practical assignment are mandatory. The final grade for the course is determined by calculating the weighted average. However, the exam grade must be above the passing mark of 5.5. Students have the opportunity to retake the exam but not the assignment.

Important Information

1. The schedule is available in [MyTimetable](#).
2. The course has a [page](#) on Brightspace. Make sure you are enrolled there.
3. Through MyStudyMap, you are required to **register for the courses, exams, and retakes**. You have up to 10 days for exams, and from 30 - 5 days for retakes before the exam begins.
4. If you didn't enroll in time, you can fill out an online registration form by sending an email to the **LIACS - Education Office** <education-office@liacs.leidenuniv.nl>.
5. Do not forget to **deregister up to 8 days before an exam session** in case you won't take part in it!
6. Lectures will be recorded – use recordings wisely!



Course Schedule (Tentative)

Feb. 4	Introduction / Biological background	Apr. 1	Practical session on Optimization and Back-propagation
Feb. 11	The Hopfield Model	Apr. 8	Convolutional NNs and Computer Vision
Feb. 18	The Hopfield Model (Part II)	Apr. 15	Practical session on Convolutional NNs
Feb. 25	Exercise Class on Hopfield Model	Apr. 22	Student Lecture (?)
March 4	Perceptron and Multi-layer NNs	Apr. 29	Recurrent NNs
March 11	Back-propagation	May 6	Practical session on Recurrent NNs
March 18	Exercise Class on Perceptron and Backpropagation	May 13	Generative models and Q&A
March 25	No Class	May 20	Optional

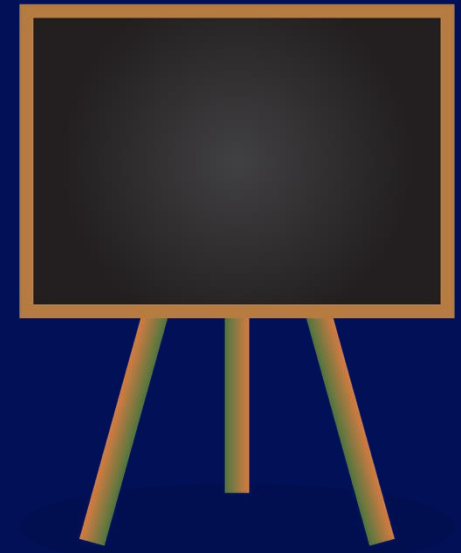
Reading List

The following titles are recommended but not mandatory for the course:

1. Hertz, John, Anders Krogh, and Richard G. Palmer. Introduction to the theory of neural computation. CRC Press, 2018.
2. Bishop, Christopher M, and Hugh Bishop. Deep Learning, Foundations and Concepts. Springer, 2023. <https://www.bishopbook.com>
3. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT press, 2016. <https://www.deeplearningbook.org/>
4. MacKay, David JC, and David JC Mac Kay. Information theory, inference and learning algorithms. Cambridge university press, 2003. <https://www.inference.org.uk/itprnn/book.pdf> (Only Chapter 5 is relevant to the course. This is very good in the learning theory, which contains various advanced and insightful discussions)

Conceptual understanding on how and why Neural Networks (NNs) work

(Blackboard)



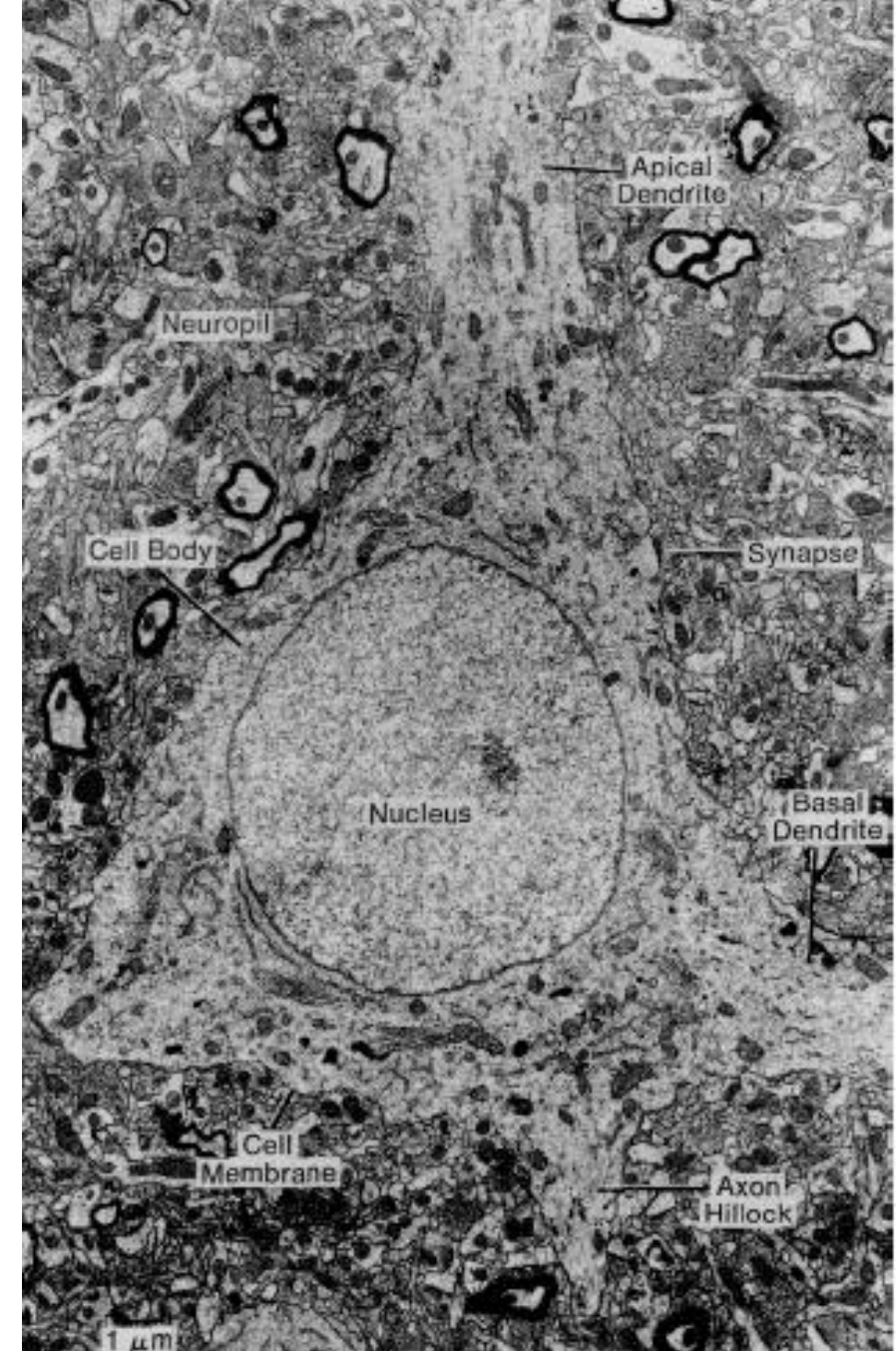
Biological Background

The Neuron

A cell that transforms and transmits information through electrical and chemical signals.

It is:

- Electrically excitable (inhibited)
- Interconnected via synapses
- Rapidly propagates signals over large distances to other cells

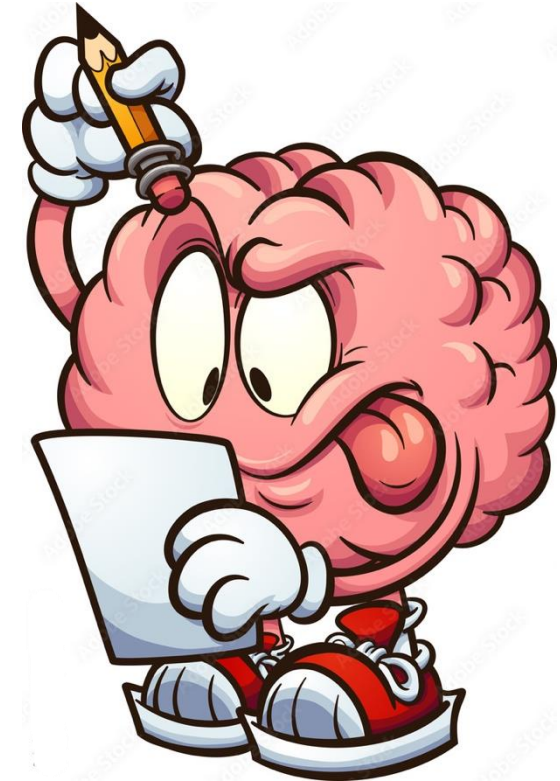


Our Brain

Our brain is:

- Robust and fault tolerant (cell die and overall performance stays the same)
- Flexible (it adjusts to new environments by learning)
- Deals with fuzzy/noisy/probabilistic information
- Highly parallel
- Small, compact, dissipates little power

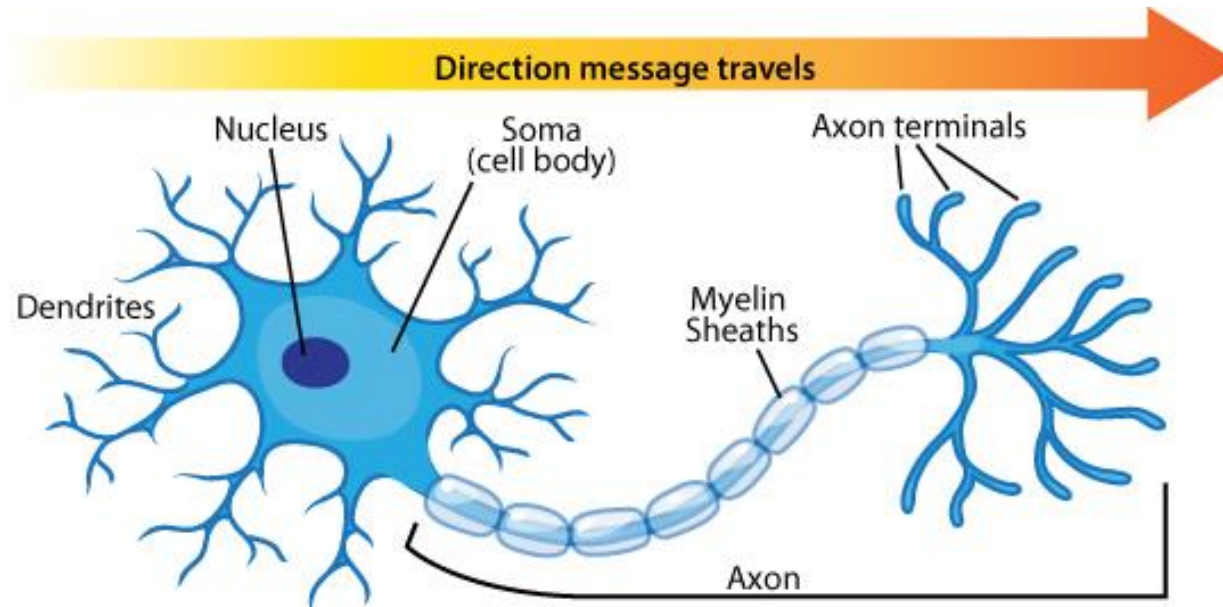
→ All these features are desirable in Artificial Systems!



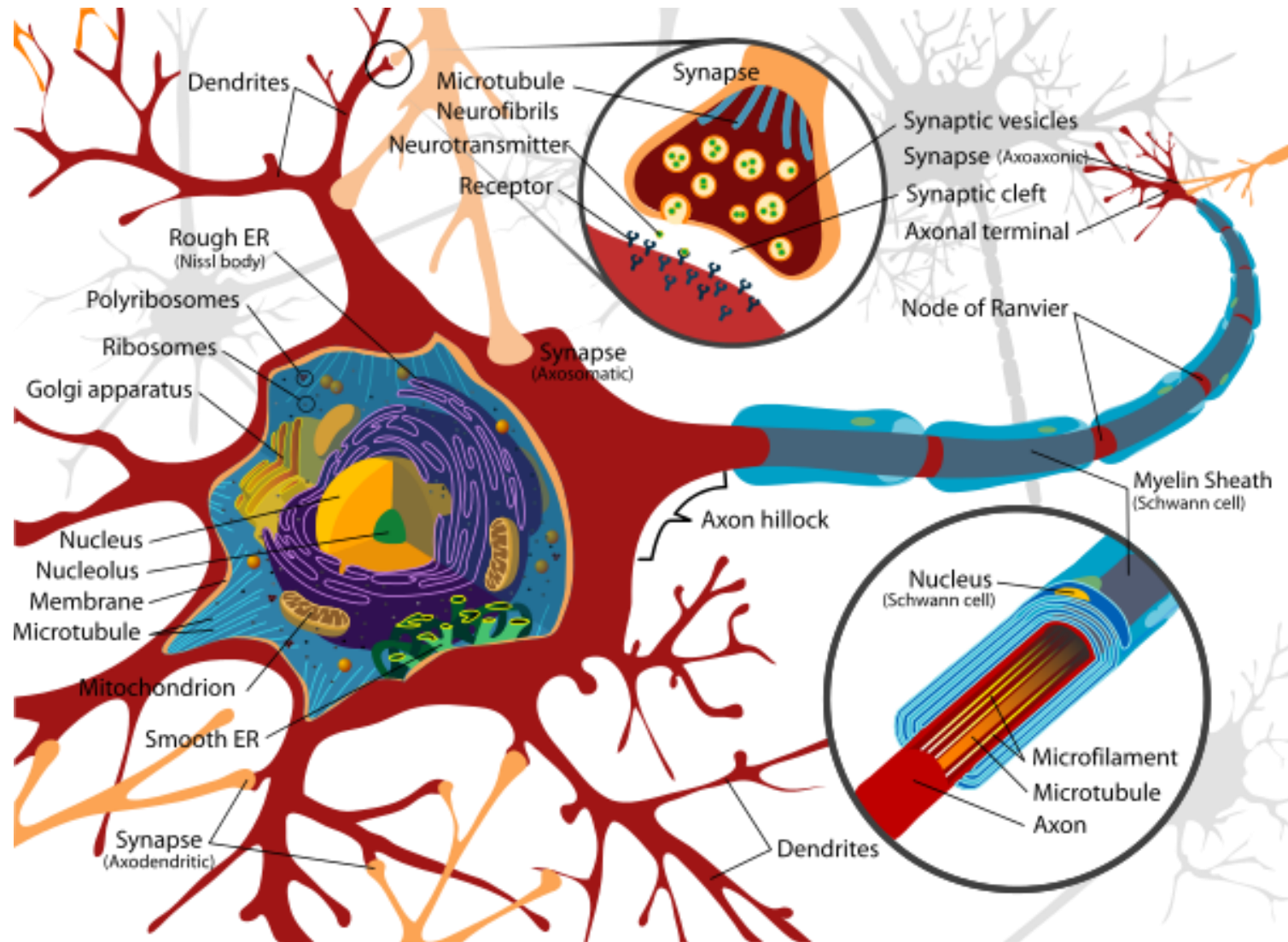
Neuron Morphology

Basic structure (that we concern):

- Dendrites: receive inputs from other neurons.
- Axon: carries the neuronal output to other cells.
- Soma: cell body, containing cell nucleus.
- Synapses: connection point between two neurons.



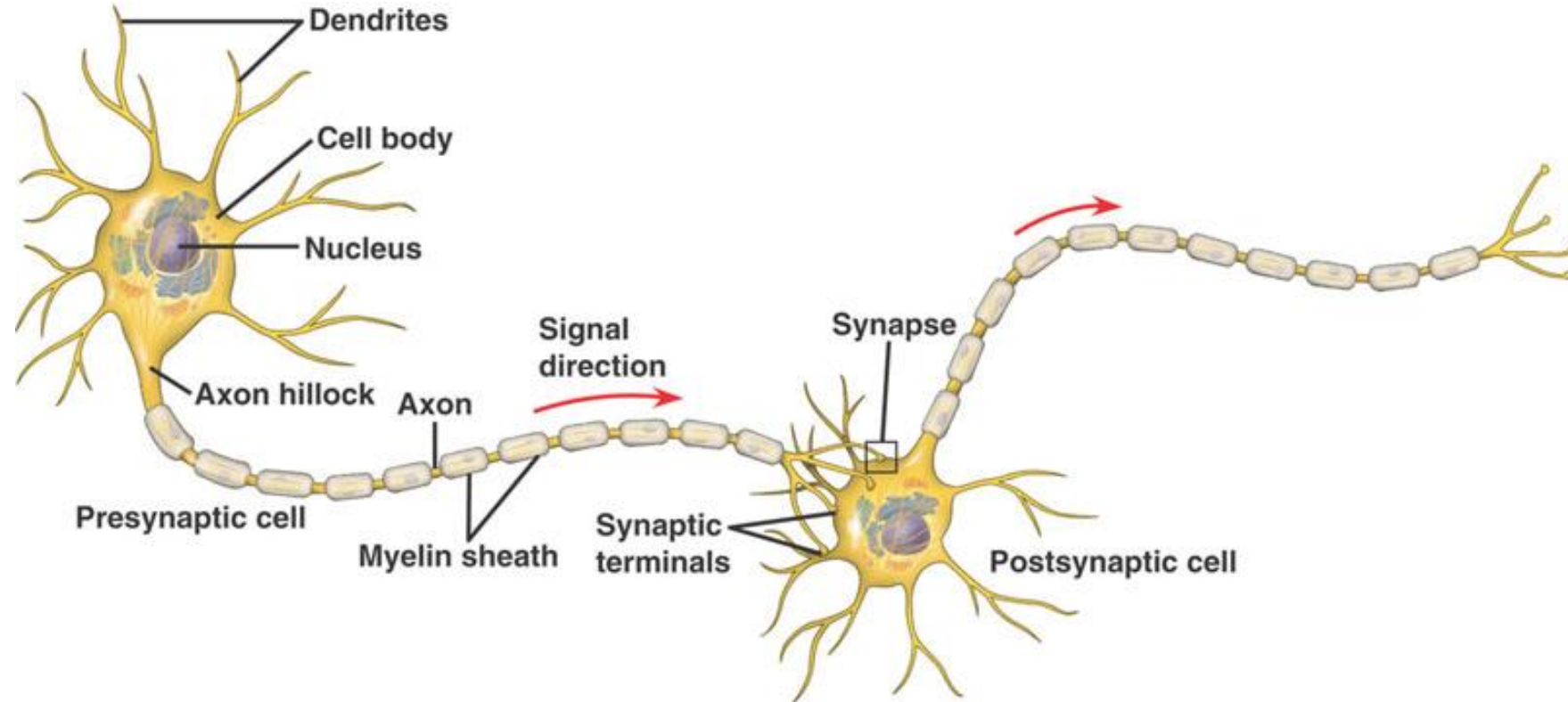
Neuron Morphology



Connections between Neurons

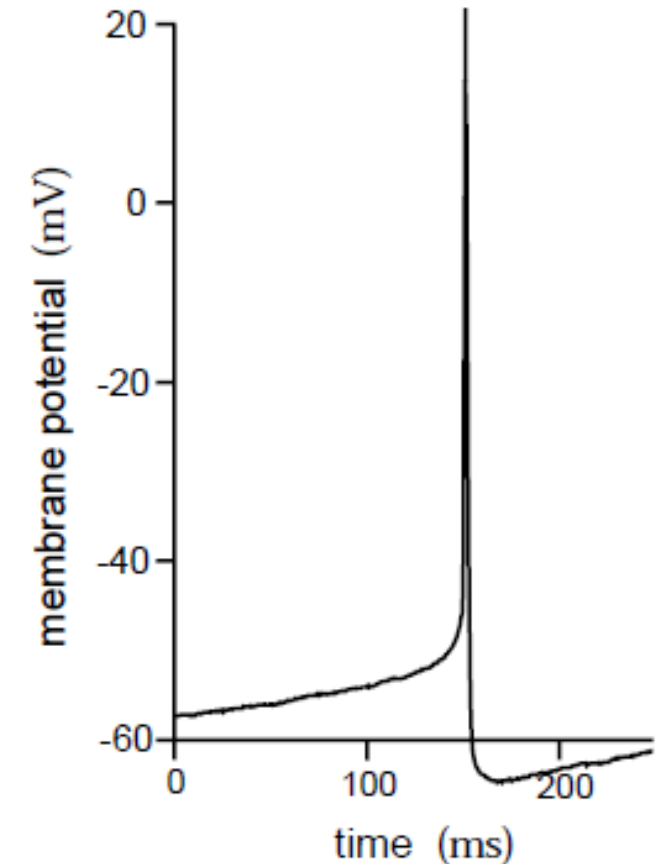
Action potential flow:

- Axon → Axon terminal → Synapses → Dendrites
- presynaptic, postsynaptic



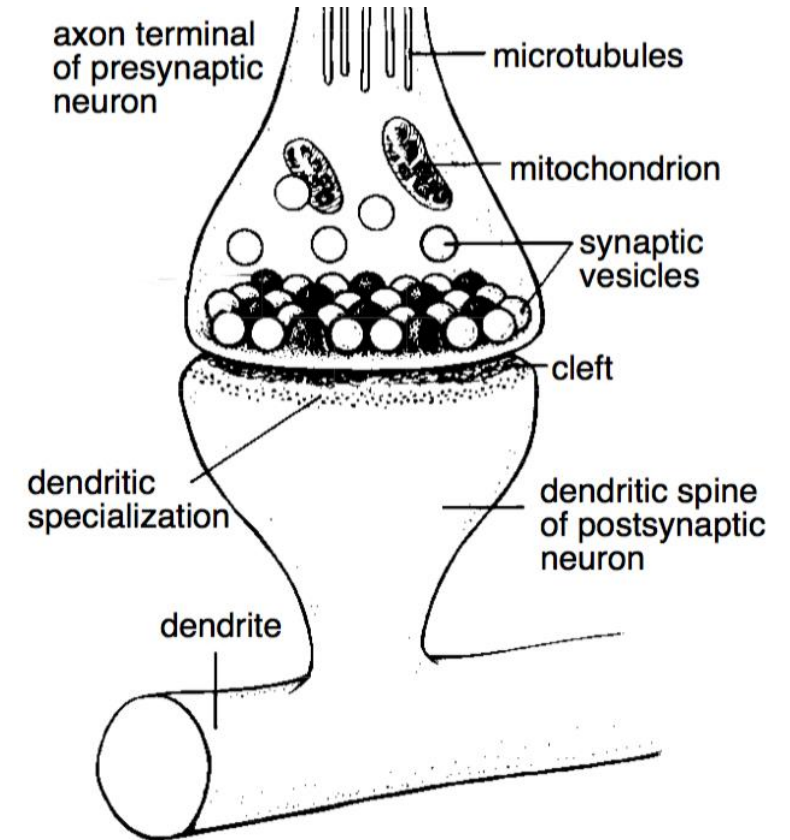
Physical Features

- Ion channels on the membrane
 - Pumps of ions
 - Na^+ , K^+ , Ca^{2+} , Cl^- (sodium, potassium, calcium, chloride)
- Membrane potential
 - Difference in electrical potential between the interior of a neuron and the surrounding extracellular medium.
 - Depolarization (sodium Na^+ ions flow inside the cell, making membrane potential less negative) -- **excitatory**
 - Hyperpolarization (potassium channels K^+ open up, letting K^+ ions to flow out the cell, making membrane potential more negative) -- **inhibitory**
- Action potential (Spike)
 - The only signal that can propagate in long distance
 - Carries information



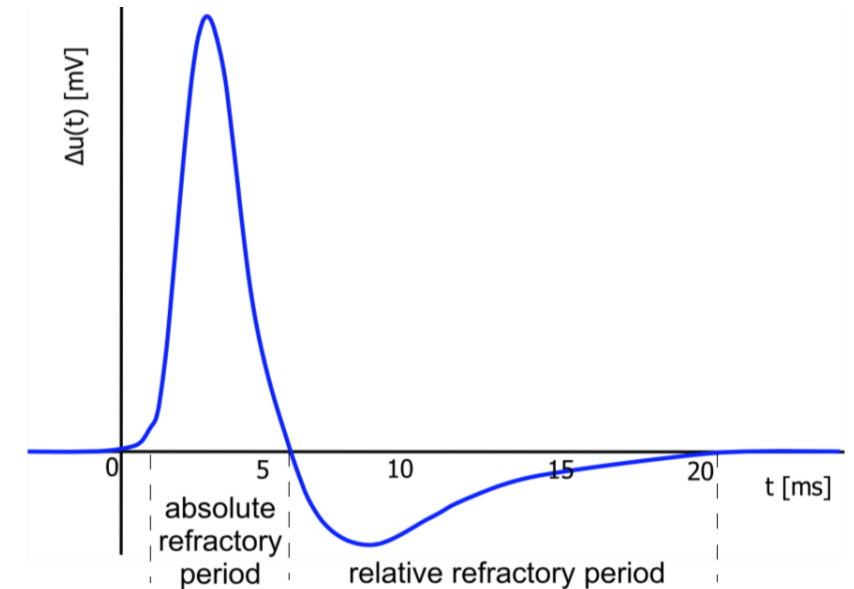
Neurotransmission

- Action potential causes the release of neurotransmitter
- The neurotransmitter binds to the receptors (on dendrites) of the postsynaptic neuron, opening ion channel.
 - Excitatory postsynaptic potential (EPSP)
 - Inhibitory postsynaptic potential (IPSP)



Elements of Neural Dynamics

- Absolute refractory period
 - Minimal temporal distance between two spikes
- Relative refractory period
 - where it is difficult, but not impossible to excite an action potential
- Negative Spike-after-potential

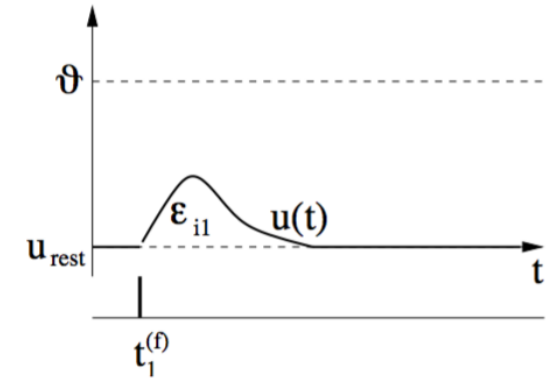
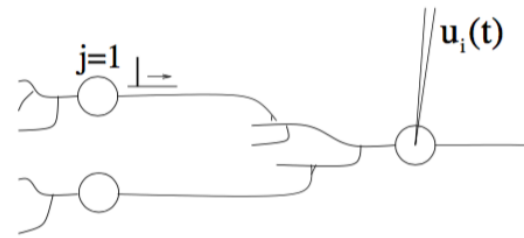


Dynamics of spike firing

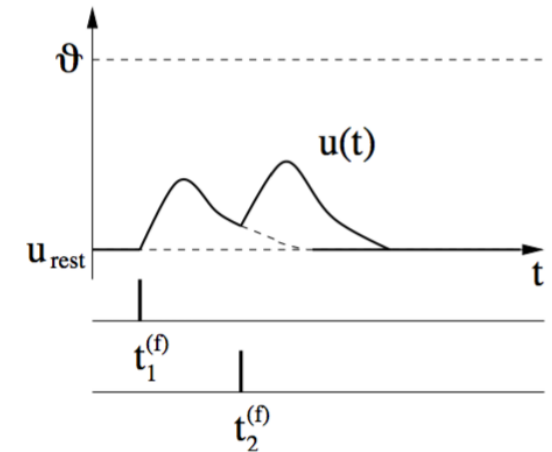
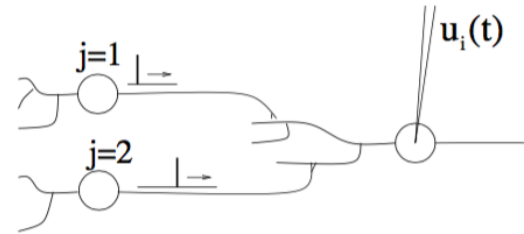
Elements of Neural Dynamics

- Postsynaptic potential (PSP)
 - Transient impact of incoming spikes
- Attenuates over time
- Additive in this model

A



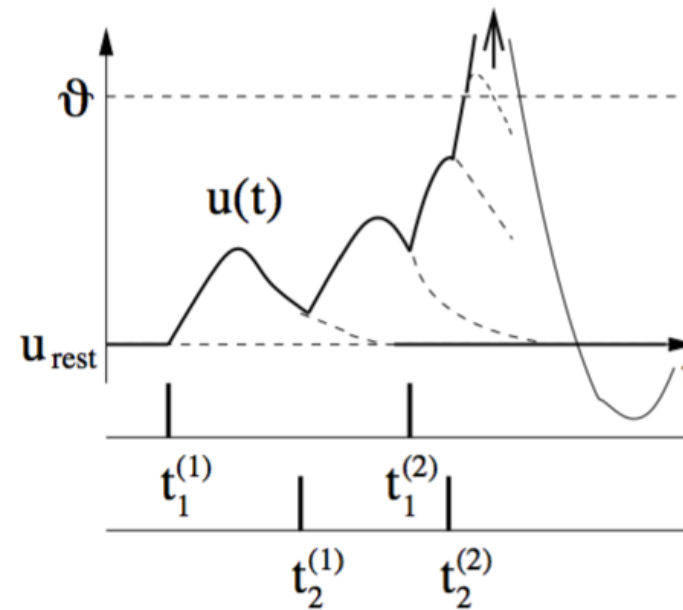
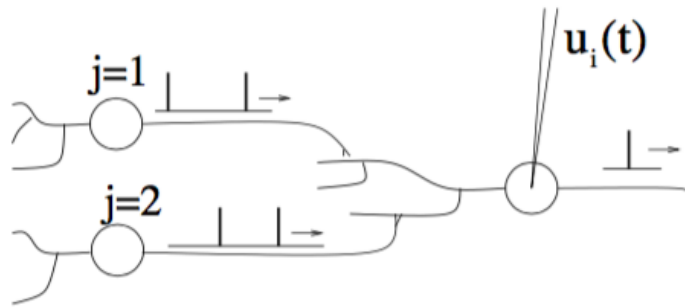
B



Elements of Neural Dynamics

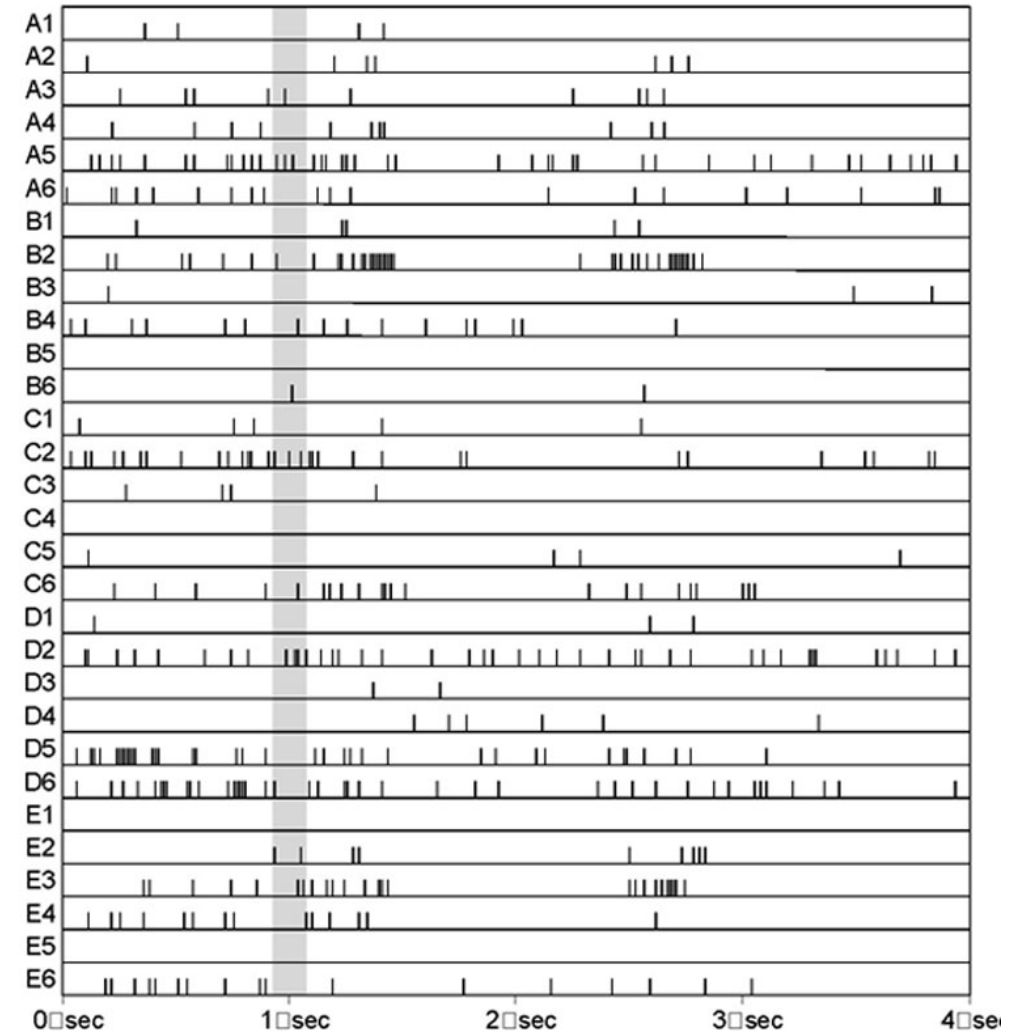
- A spike is generated if the cumulated membrane potential reaches the threshold from below
- Generalization of threshold unit

C



Spike Train

- Action potentials convey information through their timing or frequency
- Spikes → Abstraction from action potential:
 - ignore the brief duration of an action potential.
 - ignore the shape and amplitudes.
 - just keep track on when neurons are fired.
- High-dimensional time series



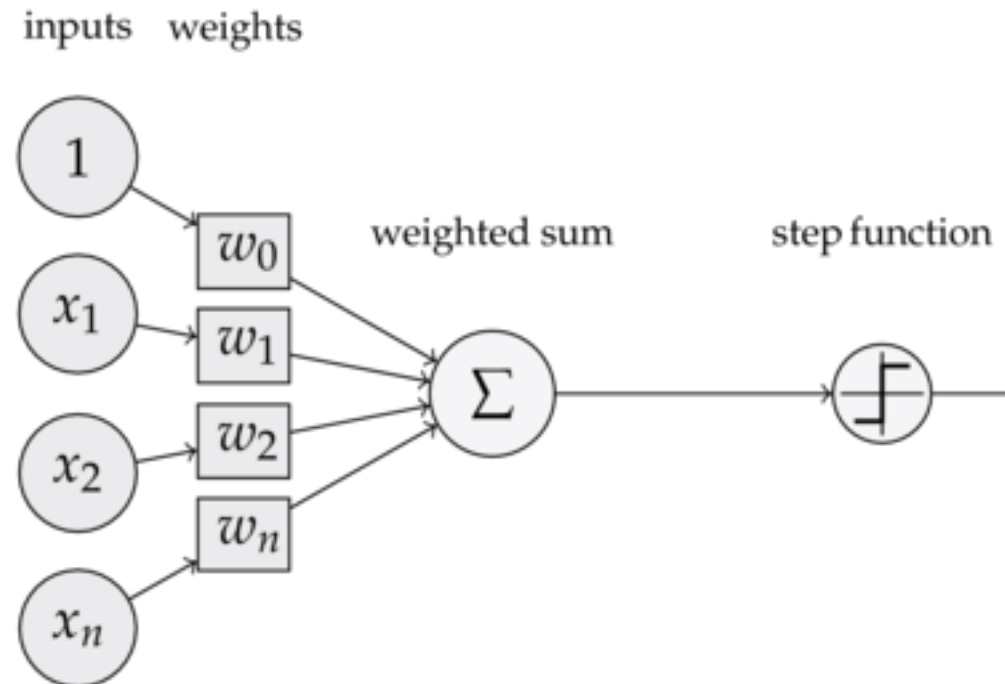
Spike trains from 30 neurons in the visual cortex of a monkey.

Some Examples of Neural Networks

Artificial Neuron

- *McCulloch-Pitts* model:

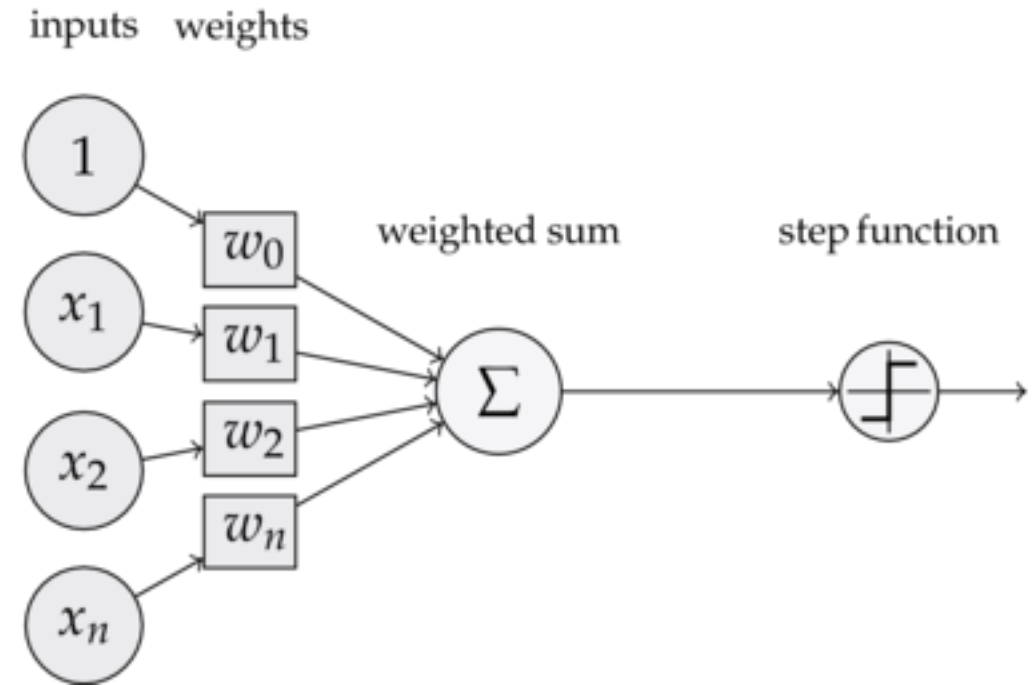
$$O = \operatorname{sgn}\left(\sum_i w_i x_i\right), \quad \operatorname{sgn}(u) = \begin{cases} 1 & \text{if } u \geq 0 \\ -1 & \text{if } u < 0 \end{cases}$$



Artificial Neuron

The McCulloch-Pitts model:

- **spikes** are interpreted as spike rates (firing rate);
- **synaptic strength** are translated as weights
- **excitation** means positive product between the incoming input and the corresponding synaptic weight
- **inhibition** means negative product between the incoming input and the corresponding synaptic weight

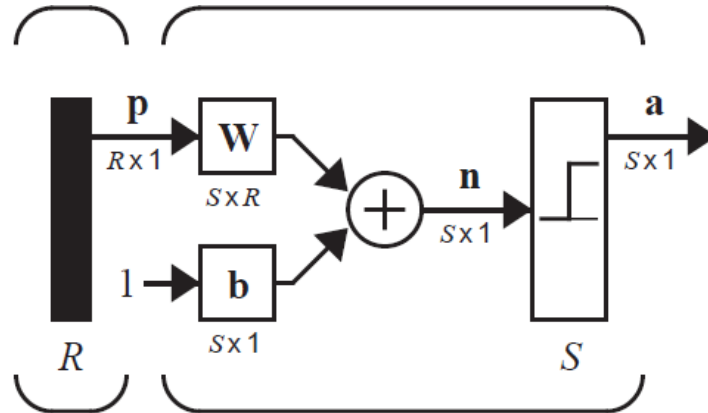


Perceptron

- Artificial neuron using a sign function (or Heaviside step function) as activation function; also called threshold unit

$$O = \text{sgn}(\vec{w}^T \vec{x} + b)$$

- It is a trainable model capable of learning from data (useful for pattern recognition and classification tasks)

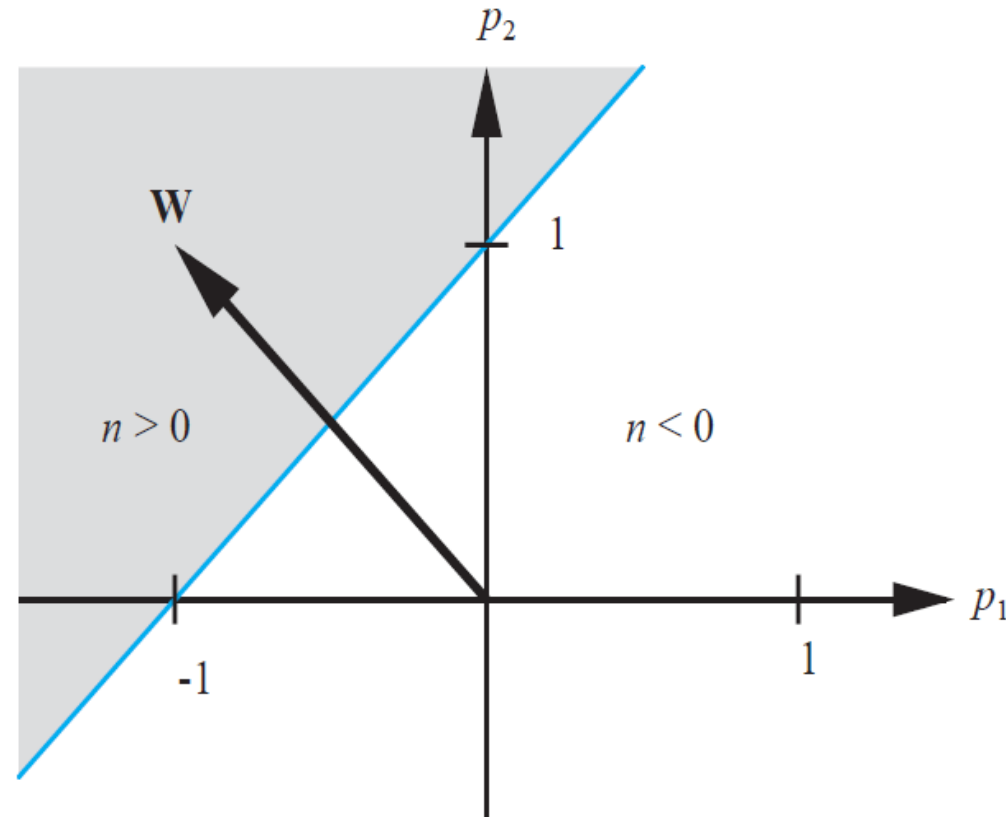


- Generalization: Replaces the threshold unit to other functions (non-linear)

Binary output $\rightarrow$ real value output

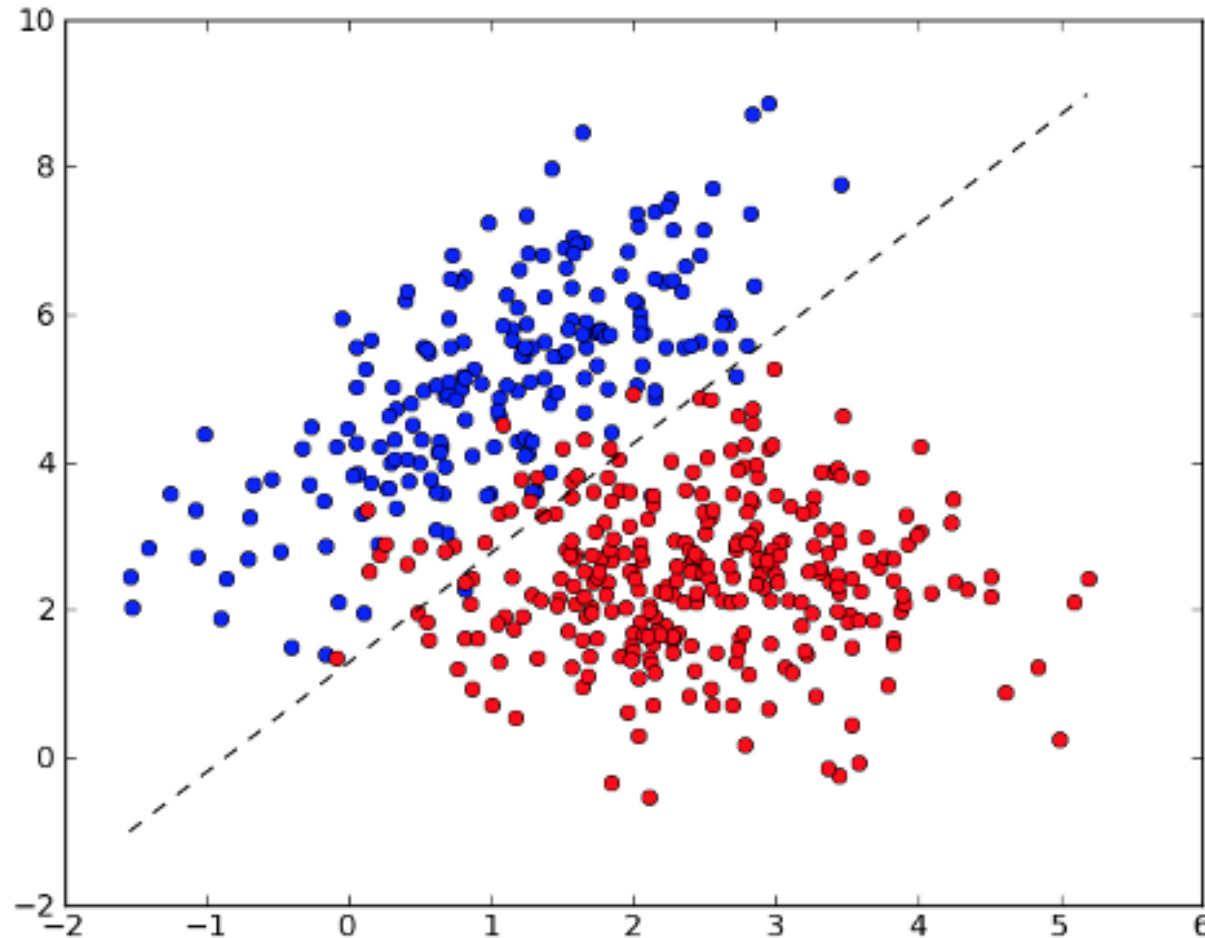
Perceptron

- Computational power: it is a linear binary classifier
- Decision boundary: $\vec{w}^T \vec{x} + b = 0$



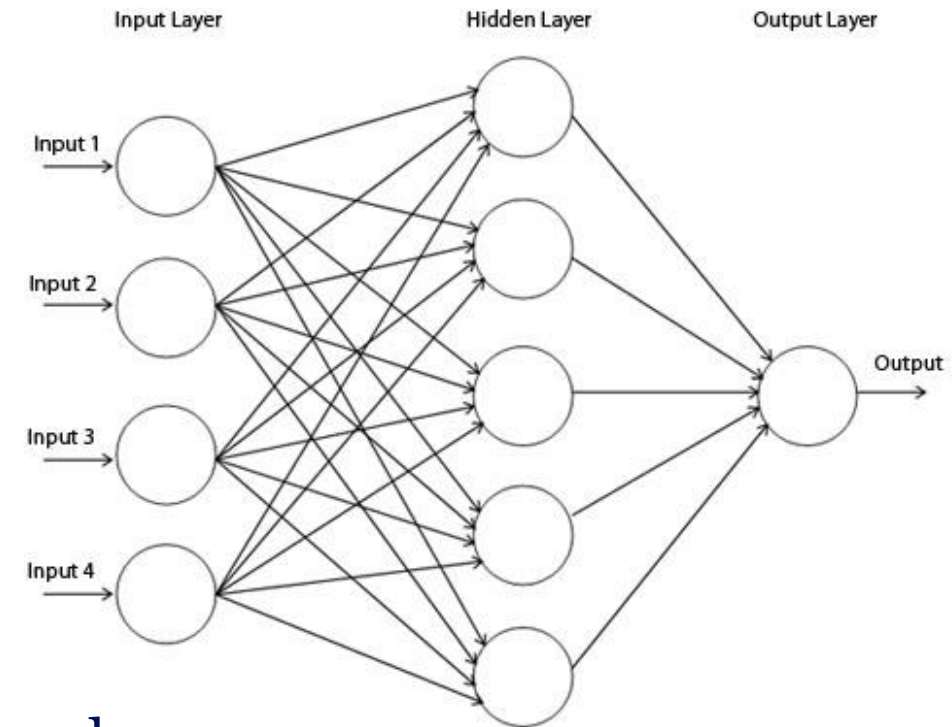
Perceptron

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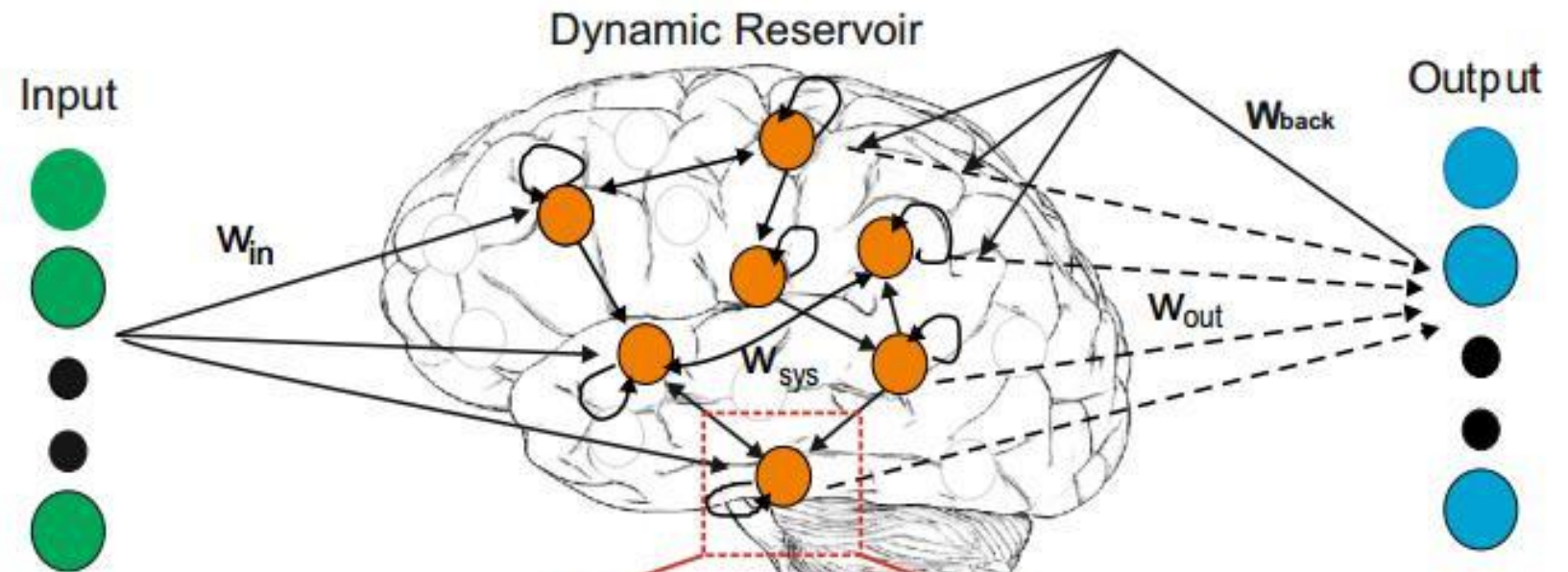
Feed forward Network

- Takes the generalized perceptron as building block
- Layered network
 - ***No cycle!***
- Network topology: connections between layers
 - Fully connected
 - Randomly connected
 - Evenly optimized → structural learning/architecture search



Recurrent Network

- Network topology: containing cycles
 - More realistic
- Dynamical system
 - Internal state of the network evolves over time
 - The Learning task is more difficult



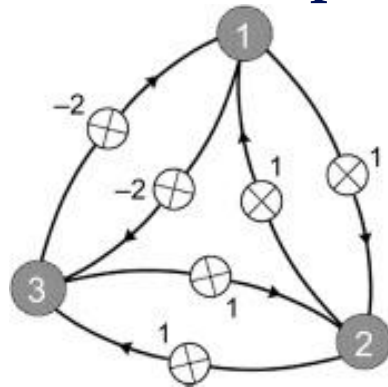
Hopfield Network

October 8, 2024

- Network equation:

$$O_i(t + 1) = \text{sgn} \left(\sum_j w_{ij} O_j(t) \right)$$

- Associative memory
 - Capable of store a collection of patterns $\{\vec{x}_i\}_{1 \leq i \leq p}$
 - If present a new (not stored) pattern $\vec{x}$, the network will produce the stored pattern most closely resemble $\vec{x}$.



John J. Hopfield, Pioneer of Artificial Neural Networks, Wins 2024 Nobel Prize in Physics (together with Geoffrey Hinton)





Thank you – Any questions?



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